Bayesian Logistic Regression for Cognitive Decline

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Overview

Instead of predicting a continuous MMSE score visit by visit, this model predict whether a patient's MMSE score falls below 24 by the end of their observation period. A score below 24 is the standard clinical cutoff for mild cognitive impairment, so this is asking whether the patient crossed into clinically meaningful threshold. We predict the binary outcome based on the patient's age and their starting MMSE score.

Data Preparation

```
con <- unz("/Users/kkmanas/ds4420-finalproject/data/NACC_visitation_data.zip",
           "investigator_ftldlbd_nacc72.csv")
df_raw <- read.csv(con)

df <- df_raw[, c("NACCID", "NACCAGE", "NACCMMSE")]

sentinel_vals <- c(88, 95, 96, 97, 98, -4)
df$NACCMMSE[df$NACCMMSE %in% sentinel_vals] <- NA
df <- df[!is.na(df$NACCMMSE) & !is.na(df$NACCAGE), ]

df <- df[order(df$NACCID, df$NACCAGE), ]

cat("Rows after cleaning:", nrow(df), "\n")
```

```
## Rows after cleaning: 99470
```

```
cat("Unique patients:    ", length(unique(df$NACCID)), "\n")
```

```
## Unique patients:     31790
```

Define Binary Outcome

For each patient, we use their baseline visit (first recorded) and their last recorded MMSE. If the last MMSE is below 24 (the clinical cutoff for mild cognitive impairment) we label that patient as impaired (1). Otherwise, 0. We also require at least 3 visits so there's a meaningful trajectory to observe.

```r
MMSE_THRESHOLD <- 24  # standard clinical cutoff for mild cognitive impairment

patient_summary <- do.call(rbind, lapply(split(df, df$NACCID), function(sub) {
  if (nrow(sub) < 3) return(NULL)
  data.frame(
    NACCID         = sub$NACCID[1],
    age_baseline   = sub$NACCAGE[1],
    mmse_baseline  = sub$NACCMMSE[1],
    mmse_last      = sub$NACCMMSE[nrow(sub)],
    n_visits       = nrow(sub)
  )
}))

# Binary label: last MMSE fell below clinical impairment threshold
patient_summary$impaired <- as.integer(
  patient_summary$mmse_last < MMSE_THRESHOLD
)

cat("Total patients:        ", nrow(patient_summary), "\n")
```

```
## Total patients:         14780
```

```r
cat("Impaired (label=1):    ", sum(patient_summary$impaired), "\n")
```

```
## Impaired (label=1):     4685
```

```r
cat("Not impaired (label=0):", sum(patient_summary$impaired == 0), "\n")
```

```
## Not impaired (label=0): 10095
```

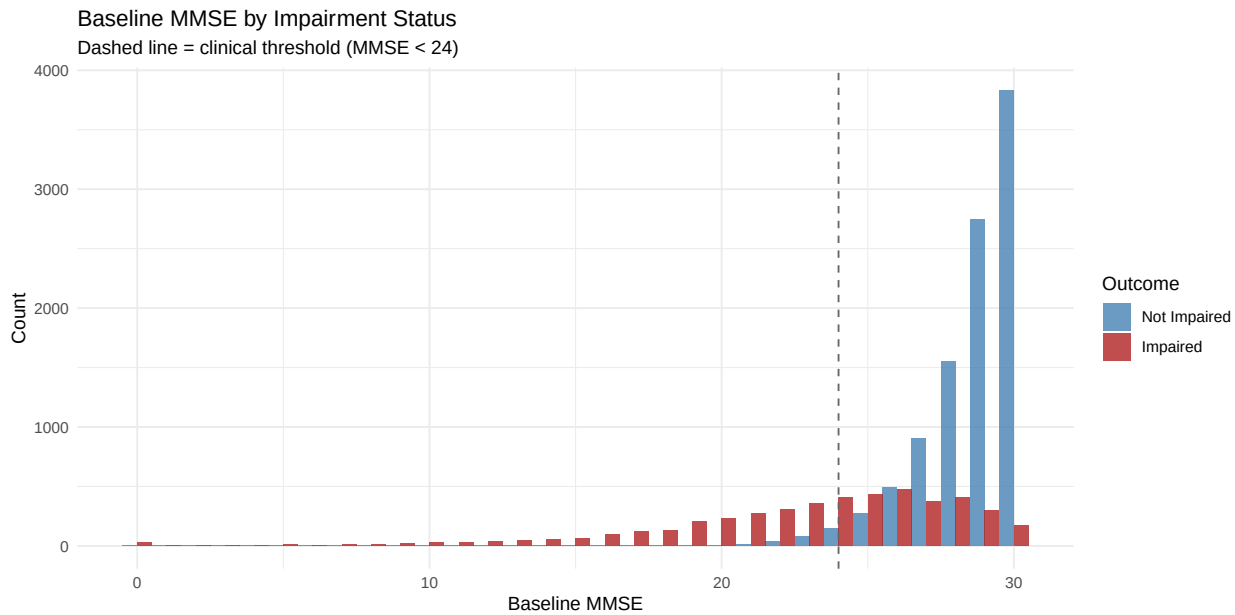
```r
cat("Impairment rate:       ", round(mean(patient_summary$impaired) * 100, 1), "%\n")
```

```
## Impairment rate:        31.7 %
```

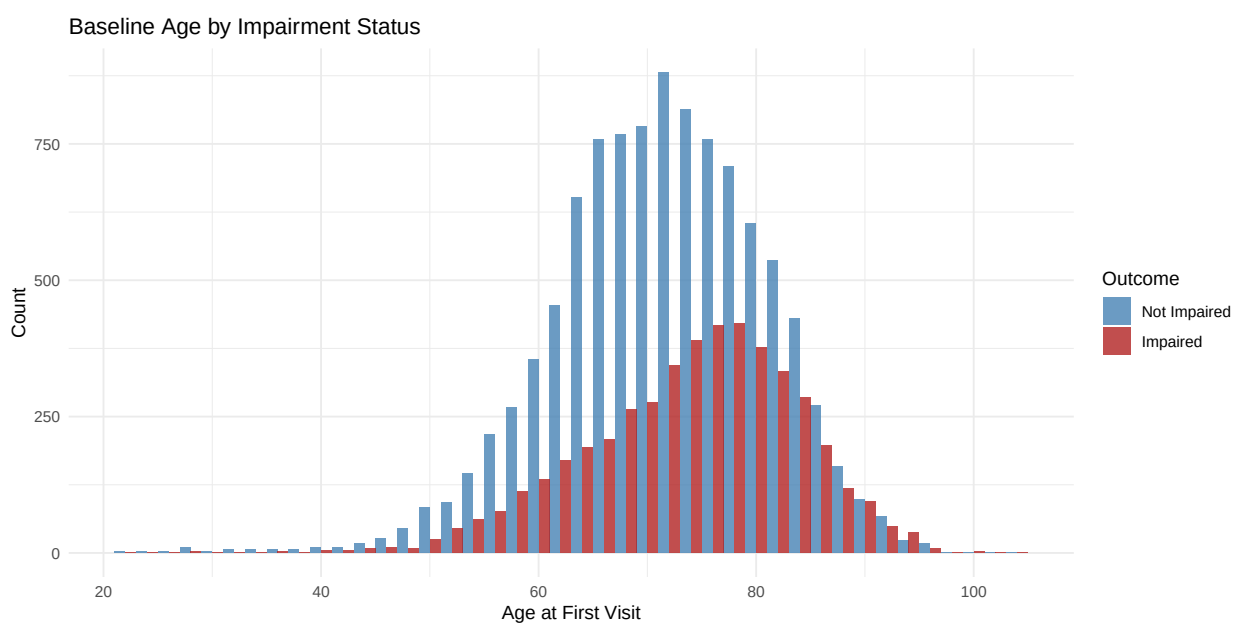
Exploratory Data Analysis

```r
library(ggplot2)

# Baseline MMSE by outcome
ggplot(patient_summary, aes(x = mmse_baseline, fill = factor(impaired))) +
  geom_histogram(binwidth = 1, position = "dodge", alpha = 0.8) +
  scale_fill_manual(values = c("steelblue", "firebrick"),
                    labels = c("Not Impaired", "Impaired")) +
  geom_vline(xintercept = MMSE_THRESHOLD, linetype = "dashed", color = "gray40") +
  labs(title = "Baseline MMSE by Impairment Status",
       subtitle = "Dashed line = clinical threshold (MMSE < 24)",
       x = "Baseline MMSE", y = "Count", fill = "Outcome") +
  theme_minimal()
```




```r
# Baseline age by outcome
ggplot(patient_summary, aes(x = age_baseline, fill = factor(impaired))) +
  geom_histogram(binwidth = 2, position = "dodge", alpha = 0.8) +
  scale_fill_manual(values = c("steelblue", "firebrick"),
                    labels = c("Not Impaired", "Impaired")) +
  labs(title = "Baseline Age by Impairment Status",
       x = "Age at First Visit", y = "Count", fill = "Outcome") +
  theme_minimal()
```



Model Specification

We model the probability that a patient's MMSE eventually falls below 24 as P(impaired = 1) = sigmoid(w0 + w1 * age + w2 * mmse_baseline).

Since logistic regression doesn't have a conjugate prior, we can't do Gibbs sampling, so we are using Metropolis-Hastings sampling.

Train/Test Split

We do an 80/20 split at the patient level.

```
n          <- nrow(patient_summary)
train_idx  <- sample(seq_len(n), floor(n * 0.8))
train_df   <- patient_summary[train_idx, ]
test_df    <- patient_summary[-train_idx, ]

cat("Training patients:", nrow(train_df), "\n")
```

```
## Training patients: 11824
```

```
cat("Test patients:    ", nrow(test_df),  "\n")
```

```
## Test patients:     2956
```

Bayesian Logistic Regression via MCMC

We run Metropolis-Hastings for 10,000 iterations, discarding the first 2,000 as burn-in. This gives us a matrix of posterior samples for the three weights.

```
library(MCMCpack)

X_train <- cbind(1, train_df$age_baseline, train_df$mmse_baseline)
y_train <- train_df$impaired

LOGIT_FILE <- "/Users/kkmanas/ds4420-finalproject/bayesian/logit_samples.rds"

if (file.exists(LOGIT_FILE)) {
  logit_samples <- readRDS(LOGIT_FILE)
  cat("Loaded saved logistic regression samples.\n")
} else {
  logit_samples <- MCMClogit(
    impaired ~ age_baseline + mmse_baseline,
    data     = train_df,
    burnin   = 2000,
    mcmc     = 10000,
    b0       = 0,       # prior mean for all weights
    B0       = 0.01,    # prior precision (1/variance = 1/100, so variance=100)
    verbose  = 500
  )
  saveRDS(logit_samples, LOGIT_FILE)
  cat("Samples saved.\n")
}
```

```
## Loaded saved logistic regression samples.
```

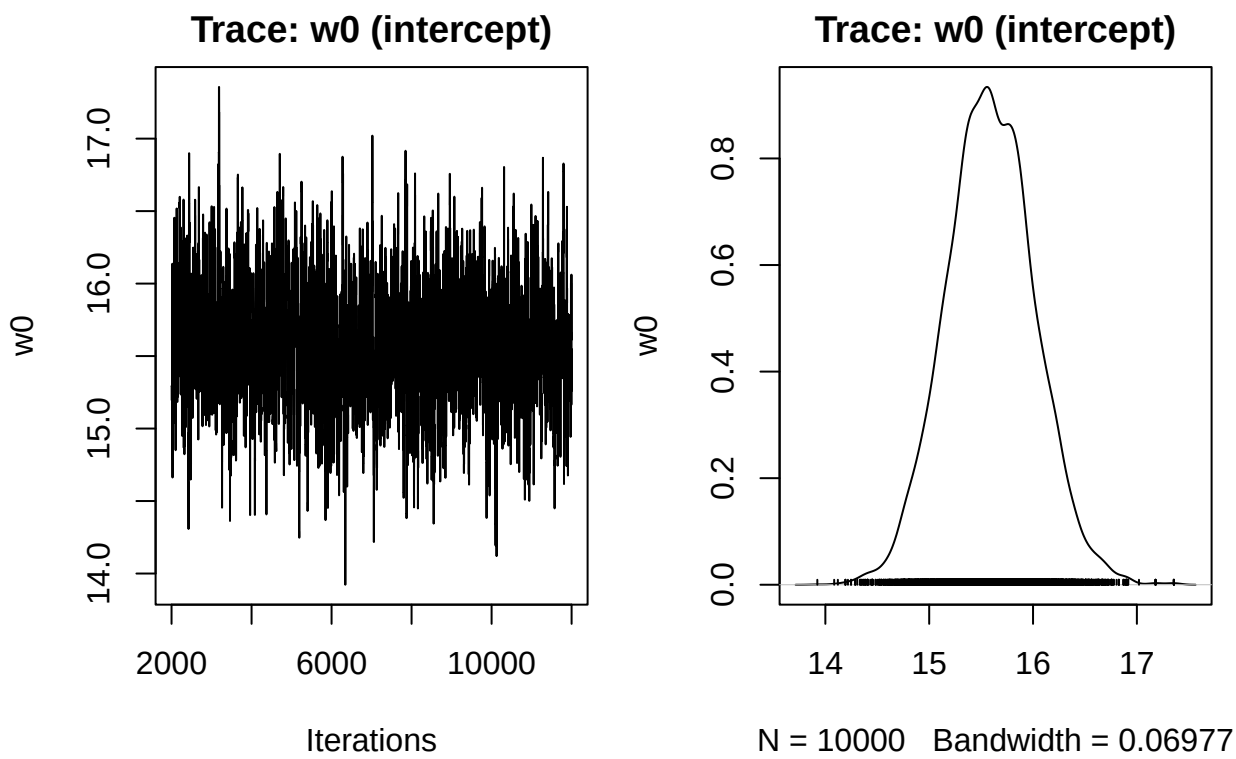
```r
cat("Posterior samples shape:", dim(logit_samples), "\n")
```

```
## Posterior samples shape: 10000 3
```

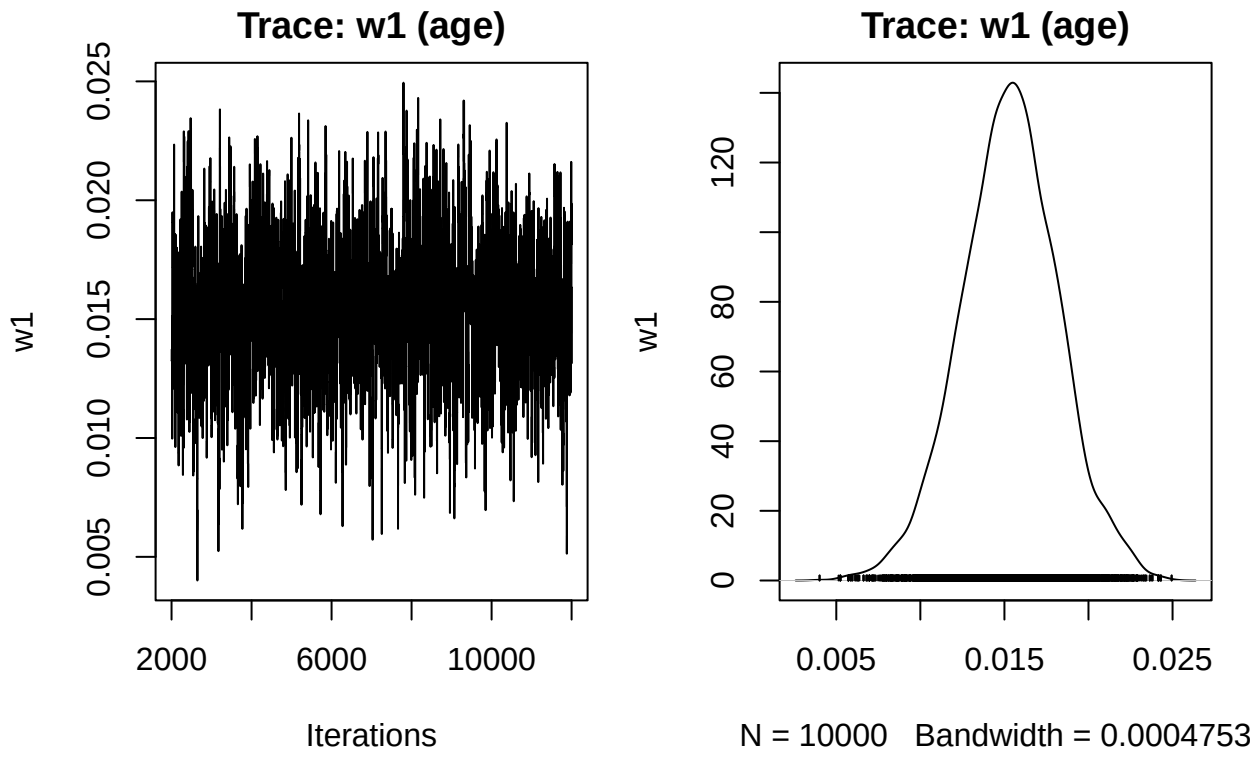
Convergence Diagnostics

Same as the linear regression model -> we check trace plots and R-hat to make sure the chain mixed well.

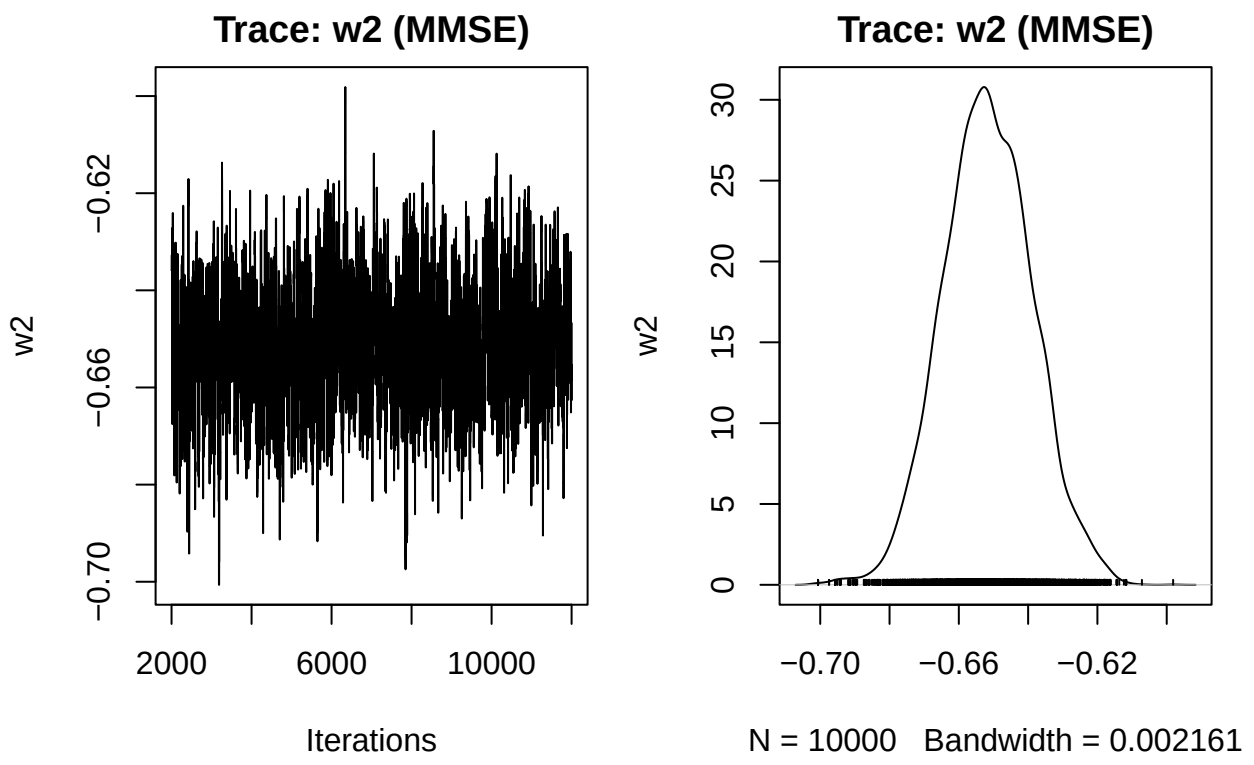
```r
par(mfrow = c(1, 3), mar = c(4, 4, 2, 1))
plot(logit_samples[, 1], type = "l", main = "Trace: w0 (intercept)", ylab = "w0")
```




```r
plot(logit_samples[, 2], type = "l", main = "Trace: w1 (age)",       ylab = "w1")
```




```
plot(logit_samples[, 3], type = "l", main = "Trace: w2 (MMSE)",      ylab = "w2")
```



```r
par(mfrow = c(1, 1))

# R-hat (split-chain)
rhat_single <- function(chain) {
  n   <- length(chain)
  h   <- floor(n / 2)
  ch1 <- chain[1:h]
  ch2 <- chain[(h + 1):(2 * h)]
  W    <- (var(ch1) + var(ch2)) / 2
  B    <- h * var(c(mean(ch1), mean(ch2)))
  Vhat <- (1 - 1/h) * W + B / h
  sqrt(Vhat / W)
}

rhats <- apply(as.matrix(logit_samples), 2, rhat_single)
cat("R-hat values:\n")
```

```
## R-hat values:
```

```r
print(round(rhats, 3))
```

```
##   (Intercept)  age_baseline mmse_baseline
##         1.001         1.001         1.000
```

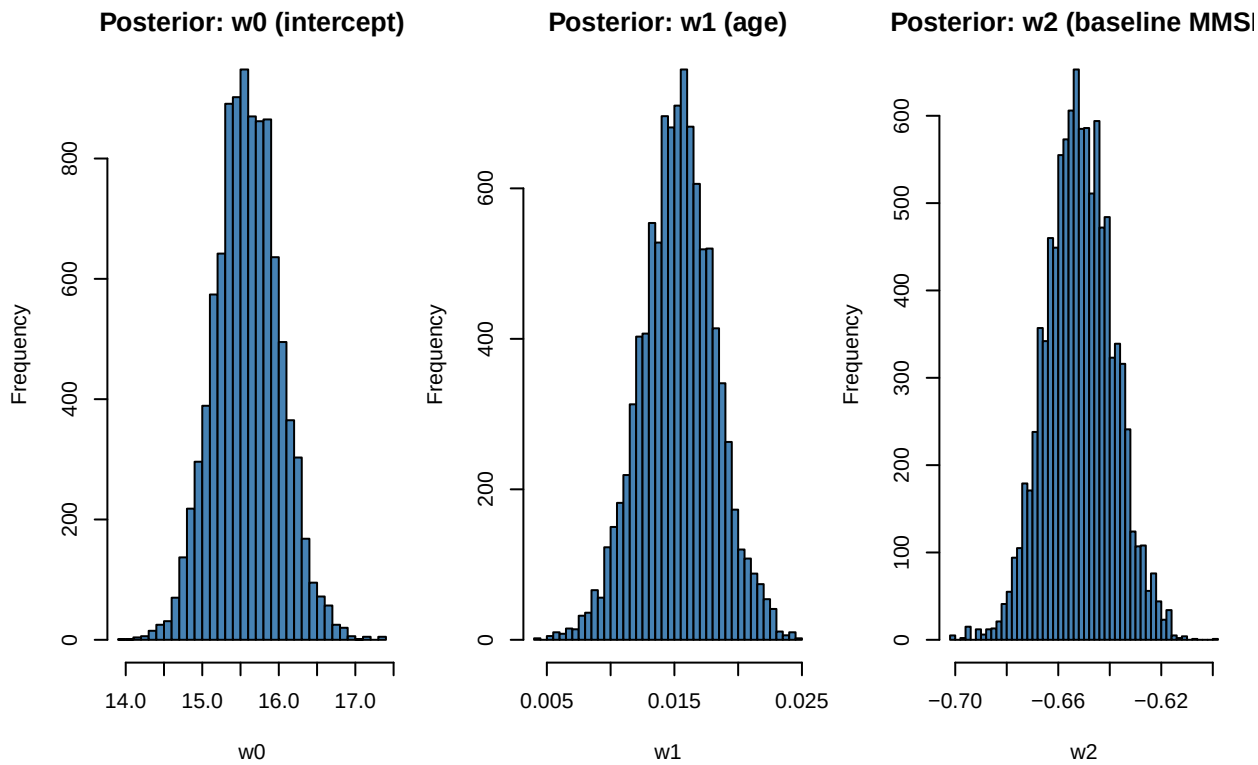
```r
cat("Max R-hat:", round(max(rhats), 3), "\n")
```

```
## Max R-hat: 1.001
```

Posterior Distributions

```r
samp_mat <- as.matrix(logit_samples)
colnames(samp_mat) <- c("w0_intercept", "w1_age", "w2_mmse")

par(mfrow = c(1, 3), mar = c(4, 4, 2, 1))
hist(samp_mat[, "w0_intercept"], breaks = 40, col = "steelblue",
     main = "Posterior: w0 (intercept)", xlab = "w0")
hist(samp_mat[, "w1_age"], breaks = 40, col = "steelblue",
     main = "Posterior: w1 (age)", xlab = "w1")
abline(v = 0, col = "red", lty = 2)
hist(samp_mat[, "w2_mmse"], breaks = 40, col = "steelblue",
     main = "Posterior: w2 (baseline MMSE)", xlab = "w2")
abline(v = 0, col = "red", lty = 2)
```




```r
par(mfrow = c(1, 1))

cat("Posterior means:\n")
```

```
## Posterior means:
```

```r
cat("  w0 (intercept):     ", round(mean(samp_mat[, "w0_intercept"]), 3), "\n")
```

```
##   w0 (intercept):      15.59
```

```r
cat("  w1 (age):           ", round(mean(samp_mat[, "w1_age"]), 3), "\n")
```

```
##   w1 (age):            0.015
```

```r
cat("  w2 (baseline MMSE): ", round(mean(samp_mat[, "w2_mmse"]), 3), "\n")
```

```
##   w2 (baseline MMSE):  -0.652
```

```r
cat("\n89% Credible Intervals:\n")
```

```
##
## 89% Credible Intervals:
```

```r
for (j in 1:3) {
  ci <- quantile(samp_mat[, j], c(0.055, 0.945))
  cat(" ", colnames(samp_mat)[j], ":", round(ci, 3), "\n")
}
```

```
##   w0_intercept : 14.919 16.264
##   w1_age : 0.011 0.02
##   w2_mmse : -0.672 -0.631
```

Predictions on Test Set

For each test patient, we find the posterior probability of decline by averaging the sigmoid of the linear predictor across all posterior samples. This gives us the probability distribution of possible outcomes.

```r
sigmoid <- function(x) 1 / (1 + exp(-x))

X_test <- cbind(1, test_df$age_baseline, test_df$mmse_baseline)

# Find predicted probability for each test patient (for each posterior sample)
pp_mat <- sigmoid(samp_mat %*% t(X_test))   # (n_samples x n_test)

# Calculate the mean
pred_probs <- colMeans(pp_mat)

# 89% credible interval on predicted probability
pred_lower <- apply(pp_mat, 2, quantile, 0.055)
pred_upper <- apply(pp_mat, 2, quantile, 0.945)

test_results <- data.frame(
  NACCID     = test_df$NACCID,
  true_label = test_df$impaired,
  pred_prob  = pred_probs,
  pred_lower = pred_lower,
  pred_upper = pred_upper,
  pred_label = as.integer(pred_probs > 0.5)
)

cat("Test observations:", nrow(test_results), "\n")
```

```
## Test observations: 2956
```

```r
head(test_results)
```

```
##       NACCID true_label  pred_prob pred_lower pred_upper pred_label
## 1 NACC000618          1 0.98473873 0.98190300 0.98732506          1
## 2 NACC000792          0 0.05515509 0.05131101 0.05920803          0
## 3 NACC001003          0 0.05205138 0.04835710 0.05596683          0
## 4 NACC001075          0 0.22064481 0.20414343 0.23696709          0
## 5 NACC001169          0 0.41155207 0.39645693 0.42713517          0
## 6 NACC001488          0 0.10932553 0.10308193 0.11580237          0
```

Evaluation

```r
# Accuracy
accuracy <- mean(test_results$pred_label == test_results$true_label)
cat("Accuracy:         ", round(accuracy * 100, 1), "%\n")
```

```
## Accuracy:          83.4 %
```

```r
# Confusion matrix
conf_mat <- table(Predicted = test_results$pred_label,
                  Actual    = test_results$true_label)
cat("\nConfusion Matrix:\n")
```

```
##
## Confusion Matrix:
```

```r
print(conf_mat)
```

```
##          Actual
## Predicted    0    1
##         0 1898  367
##         1  123  568
```

```r
# AUC
library(pROC)
```

```
## Type 'citation("pROC")' for a citation.

##
## Attaching package: 'pROC'

## The following objects are masked from 'package:stats':
##
##     cov, smooth, var
```

```r
roc_obj <- roc(test_results$true_label, test_results$pred_prob, quiet = TRUE)
cat("\nAUC:", round(auc(roc_obj), 3), "\n")
```

```
##
## AUC: 0.888
```

```r
# mean predicted prob vs actual rate by intervals of 10
test_results$decile <- cut(test_results$pred_prob, breaks = 10)
calib <- aggregate(cbind(pred_prob, true_label) ~ decile, data = test_results, mean)
cat("\nCalibration (predicted vs actual):\n")
```

```
##
## Calibration (predicted vs actual):
```

```r
print(round(calib[, c("pred_prob", "true_label")], 3))
```

```
##    pred_prob true_label
## 1      0.074      0.066
## 2      0.178      0.212
## 3      0.286      0.342
## 4      0.367      0.449
## 5      0.459      0.513
## 6      0.582      0.606
## 7      0.640      0.624
## 8      0.755      0.683
## 9      0.857      0.815
## 10     0.972      0.964
```

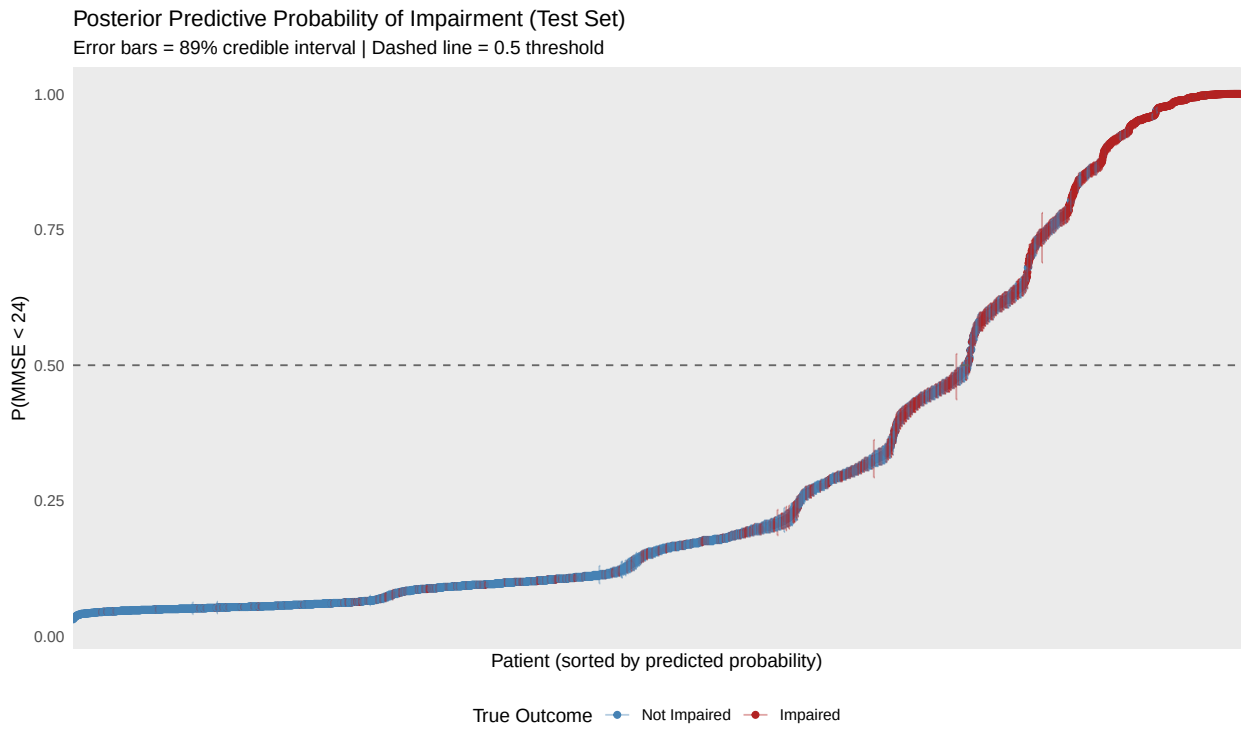
Uncertainty Visualization

This plot shows each test patient's predicted probability of decline (posterior mean).

```r
library(ggplot2)

test_results$NACCID_ordered <- reorder(test_results$NACCID, test_results$pred_prob)

ggplot(test_results, aes(x = NACCID_ordered, y = pred_prob,
                         color = factor(true_label))) +
  geom_point(size = 1.5) +
  geom_errorbar(aes(ymin = pred_lower, ymax = pred_upper), width = 0, alpha = 0.4) +
  geom_hline(yintercept = 0.5, linetype = "dashed", color = "gray40") +
  scale_color_manual(values = c("steelblue", "firebrick"),
                     labels = c("Not Impaired", "Impaired")) +
  labs(title    = "Posterior Predictive Probability of Impairment (Test Set)",
       subtitle = "Error bars = 89% credible interval | Dashed line = 0.5 threshold",
       x        = "Patient (sorted by predicted probability)",
       y        = "P(MMSE < 24)",
       color    = "True Outcome") +
  theme_minimal() +
  theme(axis.text.x = element_blank(),
        axis.ticks.x = element_blank(),
        legend.position = "bottom")
```



Summary

We fit a Bayesian logistic regression model to predict whether patients' MMSE will eventually fall below 24, which is the standard cutoff for mild cognitive impairment. We did this prediction using the baseline age and their baseline MMSE score. Rather than using the Gibbs sampler, we used a Metropolis-Hastings sampler since logistic regression has no closed-form posterior.

The result gives us a posterior distribution over the three model weights, which we used to find predictive probabilities for each test patient. Compared to the linear model, this provides more actionable insight -> we can determine whether a patient is likely to enter the more impaired range based on their first few visits.